

MAT282 · FALL 2026 · MIDTERM FORM A · ANSWER KEY

Answer key

Form A

Worked solutions and marking guide · 50 points · instructor copy: do not distribute before the exam

Read this first. Sample = 10 days. Round to 1 decimal place unless told. Apply **follow-through** (error carried forward) from B1 to C2, from C2 to C4. For C2/C4 accept a, b within ± 0.05 of the printed values. Exact values: $\bar{y} = 199.0$; $S_{xx} = 96.88$; $S_{xy} = 766.20$; $S_{yy} = 13028$; $b = 7.909$; $a = 12.35$.

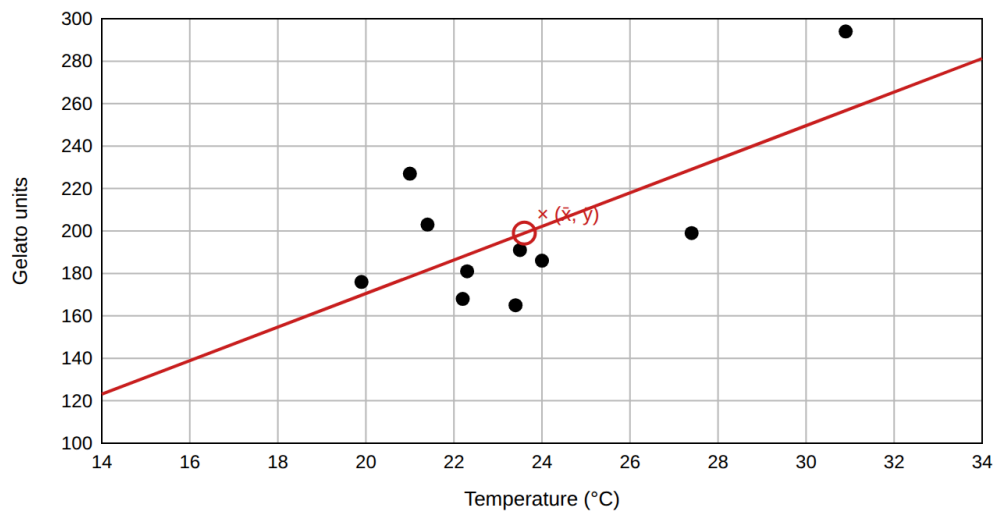
1 · Quick answer sheet

Q	Expected answer	Pts	How the marks split
A1	Direction + unit (trading day) + a falsifying result. Model: "On days with higher temperature, more gelato units are sold ($r > 0$); falsified if $r \leq 0$ or a flat cloud."	2	1 direction + unit stated; 1 falsifier that is checkable in the data
A2	Rising cloud of dots with an upward line; competing variable named with a direction (e.g. weekend $\uparrow$ sales, beach_event $\uparrow$, rain $\downarrow$).	2	1 sketch consistent with hypothesis; 1 variable + direction
A3	(a) e.g. units sold also reflect stock, price, weekend crowds, not "wanting" only; (b) reliability / comparability: same concept measured under changed conditions.	2	1 + 1
B1	Mean = $1990 \div 10 = 199.0$; median = 188.5.	2	1 each
B2	$\Sigma(\text{Units} - \text{mean})^2 = 13028$ · variance = 1447.6 · $s = 38.0$. (Dividing by 10 $\rightarrow$ 36.1: lose the variance/s marks only.)	3	1 Σ (follow-through) + 1 variance + 1 s
B3	Weekend: $n = 4$, mean 225.0 · Weekday: $n = 6$, mean 181.7.	2	1 per group
B4	$(120 \times 181.7 + 48 \times 225.0) \div 168 = 194.0$. Weekend days are over-sampled (4 of 10 = 40% vs $48/168 = 29\%$), so the plain mean is biased.	2	1 calculation + 1 reason
B5	Most extreme day: Aug 2 (294); $z = (294 - 199.0) \div 38.0 = 2.50$; $ z > 2 \rightarrow$ outlier. Any defended decision (keep and report both ways is best: day is real and consistent with hot/event day; removal needs a documented error).	2	1 z + flag; 1 justified decision
C1	10 dots within one minor grid square; $\times$ at (23.6, 199); straight line through $\sim \times$, balanced above/below, positive slope.	3	1 dots; 1 $\times$; 1 line
C2	$b = 766.20 \div 96.88 = 7.91$; $a = 199.0 - 7.91 \times 23.6 = 12.4$; $\hat{y} = 12.4 + 7.91x$; one more unit of X goes with ≈ 7.9 more servings.	3	1 b + 1 a + 1 sentence
C3	$r = 766.20 \div \sqrt{96.88 \times 13028} = 0.68$; $R^2 = 0.47 \rightarrow$ about 47% of the day-to-day variation in gelato units is associated with temperature; the rest is unexplained.	2	1 r and R^2 ; 1 interpretation (not "percent correct")
C4	Apr 8: $\hat{y}$ 188.0, res -20.0; May 10: $\hat{y}$ 178.5, res 48.5; May 15: $\hat{y}$ 198.2, res -7.2; May 21: $\hat{y}$ 202.2, res -16.2. MAE = 23.0. Worst: May 10 ($ \text{res} $ 48.5).	4	1 predictions + 1 residual signs/values + 1 MAE + 1 worst day
D1	$H2 = \text{AVERAGEIFS}(E2:E11, C2:C11, 1)$ · $H3 = \text{COUNTIFS}(C2:C11, 1, D2:D11, 1)$ · $H4 = \text{FORECAST}(26, E2:E11, B2:B11)$ (= 218.0); AVERAGEIF is acceptable for H2; FORECAST argument order is x, known_y, known_x.	3	1 each
D2	(a) The ranges have different lengths (B2:B10 has 9 cells; E2:E11 has 10). (b) $=\text{CORREL}(B2:B11, E2:E11)$.	2	1 + 1
D3	(a) Row 1 is reversed: Col 1 = weekend, Col 2 = beach_event, Col 3 = temperature_c = 5.57, Col 4 = intercept 48.97. (b) Each +1 °C goes with ≈ 5.57 more servings per day, holding the other variables fixed. (c) Intercept: expected units when all X = 0 (extrapolation: not meaningful here).	3	1 mapping + 1 interpretation "holding constant" + 1 intercept
D4	(a) $t = 25.74 \div 4.27 = 6.03$; $ t > 2 \rightarrow$ significant. (b) SE of $y = 22.5$: a forecast typically misses the actual by about 23 servings; $R^2 = 0.61$.	2	1 + 1
D5	Predicted = $48.97 + \dots = 232.4$; residual = $11 - 232.4 = 10.6$; $ \text{residual} < 22.5$: an ordinary day for the model.	2	1 predicted + residual; 1 comparison/interpretation
E1	SE = 5.39; difference = 16.7; $t = 3.10 \rightarrow$ significant. Observational before/after; season/temperature changed over the same dates (add temperature, event, weekend to LINEST).	3	1 SE + 1 t and decision + 1 limitation/variable
E2	Predicted = 276; range 231 to 321. Lost sale costs more than waste (4.00 vs 1.20) $\rightarrow$ prepare toward the upper end (accept any consistent likelihood $\times$ consequence reasoning).	2	1 range + 1 reasoned choice

F1	(i) clear hypothesis with a reason; (ii) expected sign; (iii) named LINEST cells (coef, SE, t, R^2 , SE of y / residual pattern); (iv) concrete falsifier. Accept any listed variable.	4	1 each
----	--	---	--------

2 · Worked solutions

#	Date	X	Weekend	Event	Y	Y – mean	(Y – mean) ²
1	Apr 8	22.2	0	0	168	–31.0	961.0
2	Apr 18	22.3	0	0	181	–18.0	324.0
3	May 10	21	1	0	227	28.0	784.0
4	May 15	23.5	0	0	191	–8.0	64.0
5	May 19	27.4	0	0	199	0.0	0.0
6	May 21	24	0	0	186	–13.0	169.0
7	May 26	23.4	0	0	165	–34.0	1156.0
8	Aug 2	30.9	1	1	294	95.0	9025.0
9	Aug 30	21.4	1	0	203	4.0	16.0
10	Sep 14	19.9	1	1	176	–23.0	529.0
	Total				1990	0.0	13028.0



C1 key plot. Red line: $\hat{y} = 12.4 + 7.91x$. Residuals sum to ≈ 0 for the least-squares line (they do not for a by-eye line).

LINEST decode (all 168 days). Coefficients: weekend 34.16, event 25.74, temperature_c 5.57, intercept 48.97; $R^2 = 0.61$, SE of y = 22.5, F = 86.4, df = 164. Simple regression (temperature only): $R^2 = 0.34$, SE of y = 29.3.